



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SENTHILRAJA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

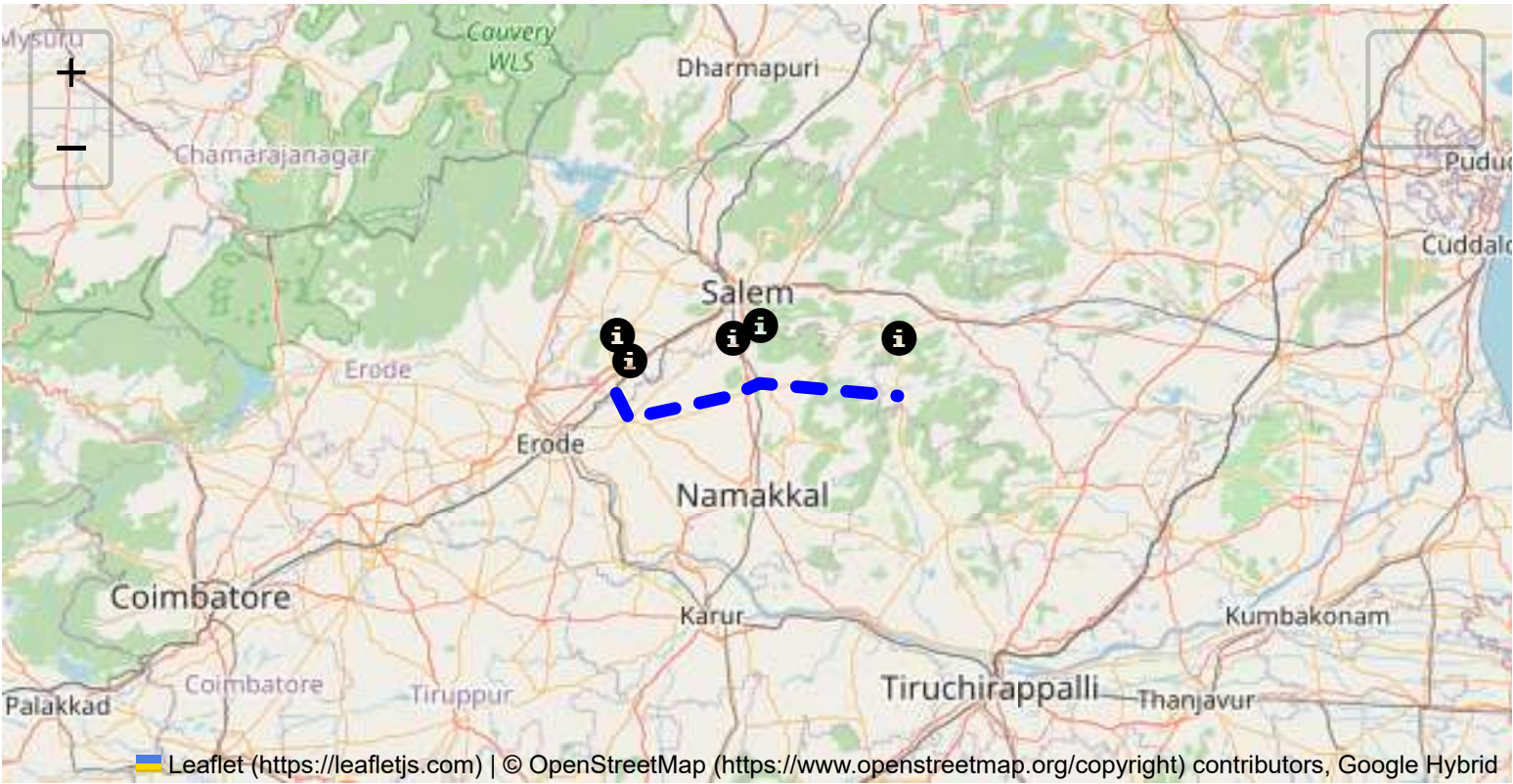
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

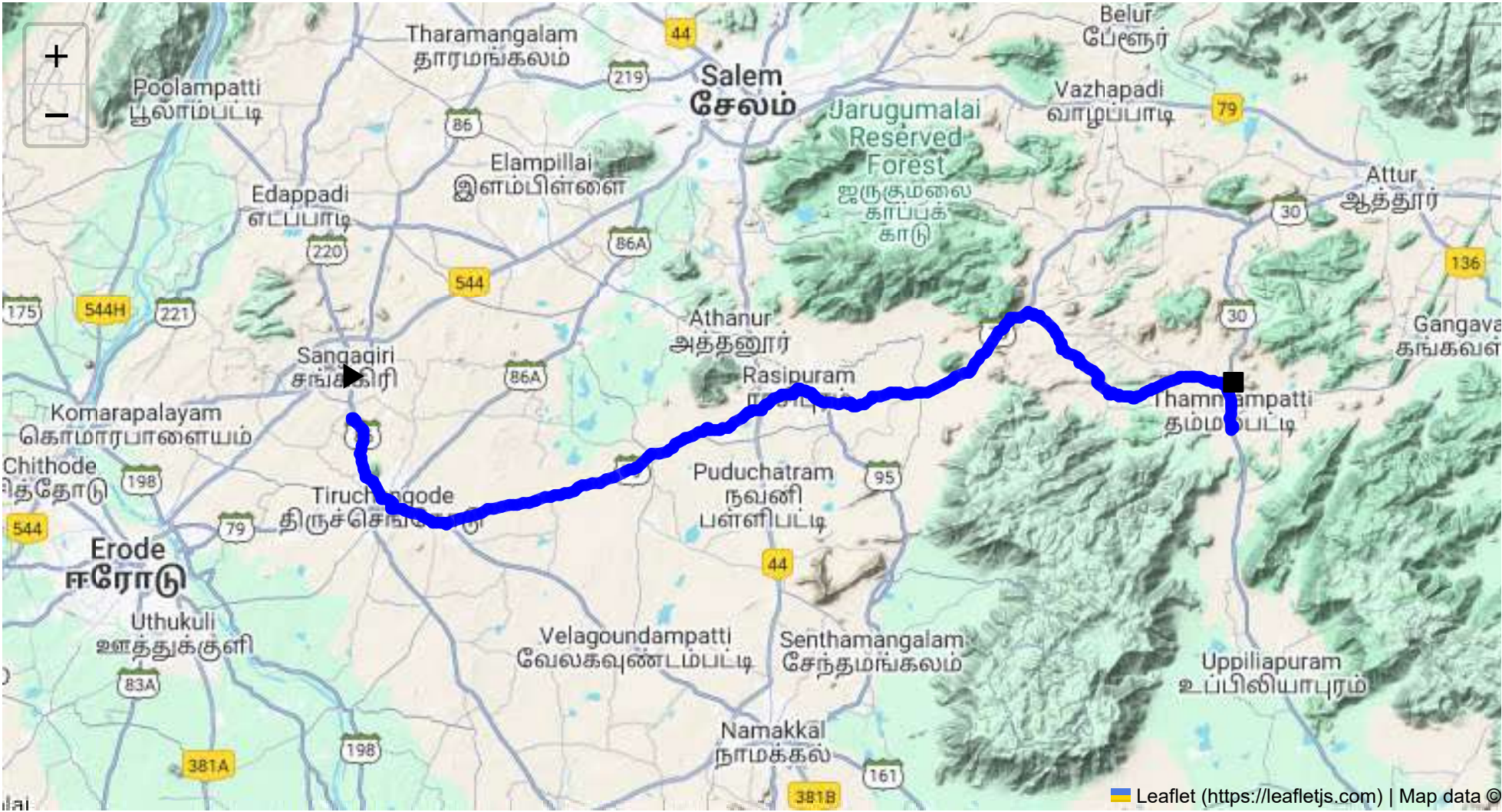
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 86.83 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.7 hours
Start: (11.4381, 77.8734)
End: (11.433693, 78.486621)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to Trichy Main Road, SH 30, spans approximately 86.83 kilometers through the regions of Sangagiri, Thokkavadi, Ponkurichi, and Rasipuram. This path primarily follows local roads and intersects with state highways such as SH 30 and SH 79.

Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate with high temperatures, especially from March to June. The monsoon season, from October to December, can result in heavy rainfall, leading to poor visibility and flooded roads. Preparations for potential landslides and waterlogged areas should be considered during these months.

Traffic Patterns

Traffic tends to be heavy during morning (7:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) peak hours, especially in and around towns like Tiruchengode and Rasipuram. Congestion is often seen near market areas and intersections with state highways.

Road Quality and Infrastructure

The road quality varies; urban sections are relatively well-maintained, while rural stretches can have potholes and uneven surfaces. Road infrastructure in town areas typically includes traffic signals and pedestrian crossings, which demand extra caution.

Alternative Routes

In emergencies, consider:

- Using NH 44 as an alternative north-south corridor if sections between Sangagiri and Rasipuram are blocked.
- Routes via SH 86 offer additional alternatives between Rasipuram and Trichy Main Road.

Local Regulations

Tamil Nadu enforces strict guidelines for the transport of hazardous materials, particularly around human habitation areas. Drivers must carry proper documentation and adhere to prescribed routes and timings, particularly avoiding transit through densely populated zones during peak hours.

Historical Incidents

While specific incident data is limited, past issues with heavy vehicles have typically revolved around mechanical failures and inadequate road conditions. No significant incidents involving hazardous materials in this specific region have been documented recently.

Environmental Considerations

Sensitive ecological areas, such as agricultural lands and water bodies, are present alongside parts of the route. Any hazardous material spill in these areas could have significant environmental impacts.

Communication Coverage

Expect reasonable communication coverage along most of the route, though some rural areas between towns may have signal dead zones, especially in hilly segments.

Emergency Response Times

- Urban centers like Sangagiri and Rasipuram can expect faster emergency response times (approximately 20-30 minutes).
- Remote areas might experience delays, with response times potentially extending to 45 minutes or more due to road conditions and accessibility.

Overall Summary of Risk Assessment

This route presents medium risk, hinging largely on weather conditions and traffic density. Preparations for adverse weather, detailed knowledge of the road conditions, and alternatives in case of emergencies are crucial for safe navigation. Ensuring compliance with local regulations is vital, especially given the environmental sensitivity along parts of the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.38346, 77.89975	30 KM/Hr
5	Turn	Medium	11.38302, 77.89995	30 KM/Hr
6	Turn	High	11.38188, 77.90257	15 KM/Hr
7	Turn	High	11.37830, 77.90160	15 KM/Hr
8	Turn	Medium	11.36754, 77.93919	30 KM/Hr
9	Turn	Medium	11.38956, 78.02898	30 KM/Hr
10	Turn	Medium	11.41279, 78.08038	30 KM/Hr
11	Turn	High	11.41463, 78.08067	15 KM/Hr
12	Turn	High	11.43248, 78.12074	15 KM/Hr
13	Turn	Medium	11.43254, 78.12131	30 KM/Hr
14	Turn	Medium	11.44497, 78.15285	30 KM/Hr
15	Turn	Medium	11.45586, 78.18051	30 KM/Hr
16	Turn	Medium	11.45612, 78.18083	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.45747, 78.18128	30 KM/Hr
18	Turn	Medium	11.46004, 78.18427	30 KM/Hr
19	Turn	Medium	11.46030, 78.18504	30 KM/Hr
20	Turn	High	11.46020, 78.18533	15 KM/Hr
21	Turn	High	11.45900, 78.18560	15 KM/Hr
22	Turn	Medium	11.45823, 78.18832	30 KM/Hr
23	Turn	Medium	11.45408, 78.19590	30 KM/Hr
24	Turn	Medium	11.44993, 78.22146	30 KM/Hr
25	Turn	Medium	11.44956, 78.22253	30 KM/Hr
26	Turn	Medium	11.44909, 78.22293	30 KM/Hr
27	Turn	High	11.44800, 78.22312	15 KM/Hr
28	Turn	Medium	11.44758, 78.22397	30 KM/Hr
29	Turn	Medium	11.49825, 78.32406	30 KM/Hr
30	Turn	Medium	11.50502, 78.33084	30 KM/Hr
31	Turn	Medium	11.50779, 78.33309	30 KM/Hr
32	Turn	Medium	11.50863, 78.33962	30 KM/Hr
33	Turn	High	11.51180, 78.34422	15 KM/Hr
34	Turn	Medium	11.51044, 78.34846	30 KM/Hr
35	Turn	Medium	11.50912, 78.35275	30 KM/Hr
36	Turn	Medium	11.48685, 78.36882	30 KM/Hr
37	Turn	Medium	11.47996, 78.38022	30 KM/Hr
38	Turn	Medium	11.46973, 78.39351	30 KM/Hr
39	Turn	Medium	11.46796, 78.39419	30 KM/Hr
40	Turn	Medium	11.46481, 78.39292	30 KM/Hr
41	Turn	Medium	11.46419, 78.39304	30 KM/Hr
42	Turn	High	11.45614, 78.40096	15 KM/Hr
43	Turn	Medium	11.45565, 78.40303	30 KM/Hr
44	Turn	Medium	11.45656, 78.40435	30 KM/Hr
45	Turn	Medium	11.45462, 78.40871	30 KM/Hr
46	Turn	Medium	11.46738, 78.45077	30 KM/Hr
47	Turn	Medium	11.46727, 78.45435	30 KM/Hr
48	Turn	Medium	11.46812, 78.45568	30 KM/Hr
49	Blind Spot	Blind Spot	11.46798, 78.45844	10 KM/Hr
50	Blind Spot	Blind Spot	11.46857, 78.45844	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
51	Turn	Medium	11.46838, 78.45957	30 KM/Hr
52	Turn	Medium	11.46802, 78.45983	30 KM/Hr
53	Turn	Medium	11.46464, 78.47392	30 KM/Hr
54	Turn	Medium	11.46421, 78.47440	30 KM/Hr
55	Turn	Medium	11.46318, 78.48057	30 KM/Hr
56	Turn	Medium	11.46218, 78.48146	30 KM/Hr
57	Turn	Medium	11.46187, 78.48264	30 KM/Hr
58	Turn	Medium	11.46109, 78.48353	30 KM/Hr
59	Blind Spot	Blind Spot	11.43106, 78.48694	10 KM/Hr
60	Blind Spot	Blind Spot	11.43373, 78.48738	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
12	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
13	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
14	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
15	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
16	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
17	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
18	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
20	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
21	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
23	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium
24	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
25	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
26	clinic	Shreetejo Ayurveda Clinic	11.444915, 78.485868	30 km/h	Medium
27	hospital	Annai Hospital	11.4437321, 78.4858654	30 km/h	Medium
28	hospital	Vinola Hospital	11.4439899, 78.4857585	30 km/h	Medium
29	hospital	The Chirist Hospital	11.4443163, 78.4862134	30 km/h	Medium
30	clinic	Sabeetha Surgical Clinic	11.4421254, 78.4867012	30 km/h	Medium
31	hospital	Dr. Ramu Life Care Hospital	11.4426079, 78.4879762	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Vidya madir school	11.4356335, 78.1381408	30 km/h	Medium
9	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
10	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
11	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
22	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38346, 77.89975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38302, 77.89995



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38188, 77.90257



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37830, 77.90160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.36754, 77.93919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38956, 78.02898



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41279, 78.08038



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41463, 78.08067



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43248, 78.12074



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43254, 78.12131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44497, 78.15285



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45586, 78.18051



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45612, 78.18083



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45747, 78.18128



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46004, 78.18427



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46030, 78.18504



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46020, 78.18533



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45900, 78.18560



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45823, 78.18832



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45408, 78.19590



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44993, 78.22146



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44956, 78.22253



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44909, 78.22293



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44800, 78.22312



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44758, 78.22397



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49825, 78.32406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50502, 78.33084



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50779, 78.33309



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50863, 78.33962



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51180, 78.34422



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51044, 78.34846



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50912, 78.35275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47996, 78.38022



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46973, 78.39351



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46796, 78.39419



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46481, 78.39292



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46419, 78.39304



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45614, 78.40096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45565, 78.40303



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 78.40435



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45462, 78.40871



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46738, 78.45077



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46727, 78.45435



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46812, 78.45568



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.46798, 78.45844



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46857, 78.45844



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46838, 78.45957



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46802, 78.45983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46464, 78.47392



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46421, 78.47440



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 78.48057



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46218, 78.48146



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46187, 78.48264



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46109, 78.48353



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.43106, 78.48694



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.43373, 78.48738

